SIRS: Multitask Joint Learning for Remote Sensing Foreground-Entity Image–Text Retrieval

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***Abstract*— The essence of improving the effect of cross-modal image–text retrieval (CIR) lies in the finer-grained modeling of homogeneous features between modalities. However, in remote sensing (RS) scenarios, existing methods usually apply the image–sentence granular feature alignment paradigm, bringing significant difficulties to the fine-grained representation of homogeneous features between modalities. Besides, more complex background noise and extreme scale ranges of foreground targets are hard to distinguish, causing the feature mottle problem. To address the above issues, we propose a novel Semantic-guided Image–text Retrieval framework with Segmentation (SIRS). It is a multitask joint learning framework for plug-and-play and end-to-end training RS CIR models efficiently, including semantic-guided spatial attention (SSA) and adaptive multiscale weighting (AMW) modules. First, SSA introduces a background reconstruction (BR) branch based on noise perception and a semantic segmentation (SS) branch based on pixel-level prediction. It explores a joint learning strategy that concisely filters background noise and refines foreground features considerably. Second, AMW performs multiscale weighting on various layers of feature map output by the encoder, effectively improving the learning efficiency of foreground targets at different scales. It is worth mentioning that SIRS outputs combination results with image and segmentation mask, which is not available in other methods. Based on the RSITMD dataset, we complete the SS annotation RSITMD-SS to verify the performance of the proposed method. Sufficient and complete experiments verify the effectiveness of the proposed method. With SIRS, the mainstream SVP and CLIP-based methods improve about 7 mR and derive segmentation prediction with acceptable computational cost optionally. The code and associated dataset will be available at https://github.com/StarBurstStream0/SIRS.**

***Index Terms*— Cross-modal image–text retrieval (CIR), feature misalignment, foreground-entity granularity, multitask learning, remote sensing (RS), semantic segmentation (SS).**

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ACRONYMS

RS	Remote sensing.
CIR	Cross-modal image–text retrieval.
SS	Semantic segmentation.
BR	Background reconstruction.
SVP	Supervised vision pretrained.
CLIP	Contrastive vision-language pretrained.
SSA	Semantic-guided spatial attention.
AMW	Adaptive multiscale weighting.
BiGRU	Bidirectional gated recurrent unit.
ResNet	Residual network.
ViT	Vision transformer.

I. INTRODUCTION

IN RECENT years, with the rapid development of RS satellite technology, RS data has made a qualitative leap in volume and quality [1], [2], [3], [4]. How to retrieve RS data swiftly becomes more and more essential. Among all retrieval forms, image–text retrieval is a classic and common cross-modal information retrieval task, and its status in RS scenarios has been increasing recently [5], [6], [7]. In recent years, RS data has demonstrated its unique characteristics [8], [9] in various modalities [10], [11], scenarios [12], [13], and tasks [14], [15], [16], pushing its efficient and accurate retrieval a challenge. Therefore, designing an image–text retrieval model that can accurately and efficiently process massive data according to the characteristics of RS scenes has become a hot topic in its field.

Current mainstream CIR can be divided into three directions. They are cross-modal feature reconstruction (CFR), cross-modal joint embedding (CJE), and feature similarity measurement (FSM). The CFR method [17], [18] regards the CIR task as a conversion between two modalities. It utilizes the deep autoencoder method to reconstruct cross-modal information twice in forward and reverse directions, effectively reducing the heterogeneous differences between modalities. The core idea of the CJE method [19], [20] is to guide the model to learn the feature differences between pairs of cross-modal data. It learns the correlation of image and text modal data through joint training, and embeds semantic features, finally realizing image–text matching. The last FSM method [21], [22], [23], [24], [25] learns the same-dimensional embedding of the input image and text, separately and succinctly. And it uses the similarity measurement to calculate these

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embeddings, deriving much higher efficiency than the first two. Considering the method scalability and the processing efficiency of massive data, we conduct our research based on the FSM method.

CIR tasks in RS scenarios mainly face the following two challenges. First, existing mainstream datasets [5], [26], [27] usually contain image–sentence-level granularity between these two modalities. However, the reasoning of human brain inquiry lies in establishing the correspondence between the entities in the sentence and the foreground objects in the image. It is tricky for a model to learn this fine-grained relationship unsupervised, especially in RS scenarios that contain massive background noise. It further causes the feature misalignment problem during the similarity calculation, leading to poor accuracy of the model. Second, RS scenes usually contain complex background noise and extreme-scale foreground objects. Although operations such as attention mechanism [28] and multiscale feature fusion [29] have emerged, their performance in CIR tasks is mediocre. We analyze that the reason is that different tasks hold different requirements for feature representation. The CIR task aims to align homogeneous features, while other vision tasks strive to model the object structure indiscriminately. To sum up, in RS scenarios, complex background information, extreme-scale foreground targets, and inappropriate feature modeling methods jointly lead to the coexistence of samples and noise in the feature space. We call these overlapped and mixed features the mottled features. They significantly increase the difficulty of aligning text modal features, affecting retrieval performance. We summarize the above phenomenon in the CIR task as a feature-mottled problem.

To address the problem of feature misalignment, we have made improvements in the data and model, respectively. On the one hand, we perform mask annotation for SS tasks based on the open-source dataset RSITMD. Specifically, we map entities appearing in text captions to foreground objects in images and annotate their outlines (see Section III-D). We name this dual-task dataset RSITMD-SS. It provides foreground-entity granular-level correspondence, laying a solid foundation for future research. To fully utilize RSITMD-SS, we propose SSA. It consists of a BR branch and a SS branch. The former guides the model to learn the distribution of background information, thereby filtering the noisy background mixed in the visual embedding. The latter refines the feature representation from the encoder by introducing the SS task of pixel-level prediction, providing more fine-grained foreground features. It effectively improves the model's ability to express fine-grained features, enriching retrieval results (see Fig. 1).

To alleviate the feature-mottled problem, we propose the AMW. AMW broadens the feature hierarchy of visual encoder output. It enriches the scale diversity of corresponding features and introduces more underlying texture information. Furthermore, it is coupled with feature similarity calculation subtly (from one-to-one to one-to-many). This change improves the fault tolerance rate of feature alignment and makes the model more stable in RS scenarios.

The main contributions of this work (see Fig. 2) can be summarized as follows.

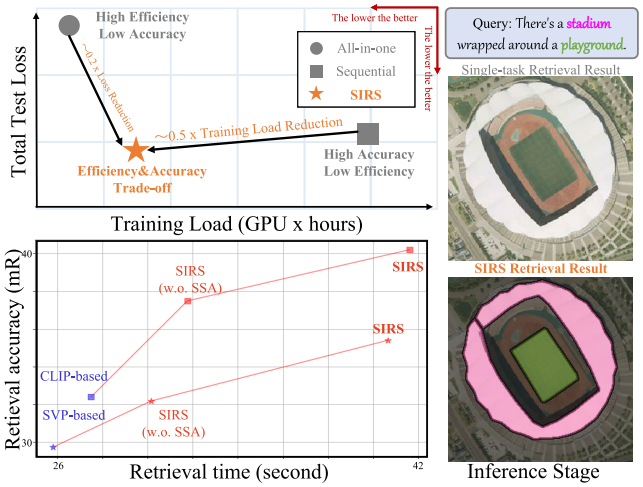


Fig. 1. Proposed SIRS achieves three advantages during training and inference: 1) in terms of training, it effectively combines the advantages of the all-in-one method and further improves the convergence of the model; 2) during inference, it improves the performance by an average of 7 mR based on both mainstream methods, with an acceptable increase in RT; and 3) it also outputs segmentation mask predictions highly relevant to the query entity.

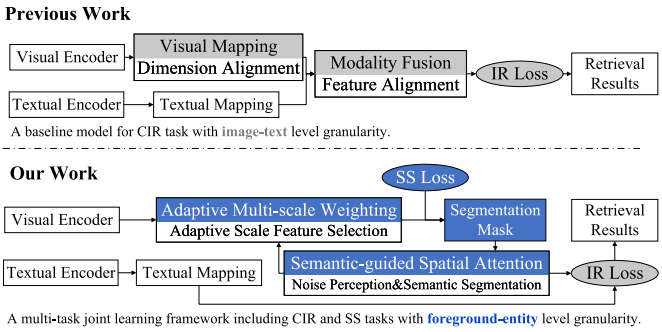


Fig. 2. Comparison between the previous work and our work.

1) We propose a semantic-guided image–text retrieval framework with segmentation (SIRS), a plug-and-play multitask joint learning framework. It aims to align the fine-grained visually homogeneous features based on semantic mask predictions in the RSCIR task.
2) SSA learns the distribution probability of background pixels based on the noise perception mechanism and introduces a segmentation branch to learn the pixel-level semantic information.
3) AMW enriches the scale diversity of encoder output features, thereby alleviating the feature-mottled problem in RS scenarios.
4) Fourteen categories of SS masks have been labeled on the RSITMD dataset. Experiments show that SIRS improves SVP and CLIP-based methods by an average of 7 mR. Besides, it derives a segmentation mask, which is unavailable for other methods.

II. Related Works

In this section, we mainly introduce three parts. They are natural scene CIR, RS CIR, and RS joint learning.

A. Natural Scene CIR

With the help of a large amount of data and elaborate annotations, many novel methods [17], [18], [19], [20] have

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emerged for the natural scene CIR task. Faghri et al. [30] proposed a novel loss based on violations incurred by relatively hard negatives inspired by structured prediction to improve performance. Lee et al. [31] presented stacked cross attention to discover the alignments between image regions and words, capturing the fine-grained interplay.

Limited by the model structure and training strategy, the above methods are unsatisfactory in model complexity and inferencing efficiency. Based on this, some methods [21], [22] embrace the feature alignment and similarity measurement strategy, improving the efficiency of model retrieval. Zhu et al. [32] proposed the external space attention to enable element-wise fusion of modal features to balance effectiveness and efficiency. Based on regional relationships and global semantic reasoning, visual semantic reasoning network (VSRN) [33] generates enhanced visual representation to capture key objects and semantic concepts. Diao et al. [34] suppressed the disturbances of less-meaning alignments through similarity graph reasoning and attention filtration (SGRAF). Qu et al. [35] first explored interaction patterns through dynamic routing learning, demonstrating its effectiveness and superiority.

B. RS Cross-Modal Image–Text Retrieval

Due to the massive amount of data and the unique imaging mechanism in the RS scene, the method based on FSM is widely adopted for its balance of accuracy and efficiency. Yuan et al. [5] proposed the first applicable dataset, RSITMD, in RS scenarios and constructed a benchmark for CIR tasks. For reasonably fusing global and local information, a graph-based GaLR [36] was designed to model salient objects in RS scenarios. In addition, a semantic alignment network [6] introduces attention and gating mechanisms to filter noise in RS scenarios. A parameter-efficient transfer learning (PETL) method based on contrastive vision-language pretraining (CLIP) models [7] was proposed to share prior knowledge of natural scenes in RS scenarios.

To sum up, the CIR task in RS scenes mainly faces challenges such as rough-grained data annotation and insufficient learning of RS characteristics.

C. RS Multitask Joint Learning

In recent years, RS multitask joint learning has gradually emerged, mainly focusing on single-modal visual tasks such as scene classification and SS. Zhang et al. [37] explored the auxiliary role of image quality assessment (IQA) in scene classification tasks, which can achieve ideal classification results even under image distortion. To further reduce the impact of redundant noise on classification results in RS scenarios, Yang et al. [38] introduced a detection branch to refine the foreground region. Guo et al. [39] designed a scene-driven building extraction network, with the help of the classification branch, to assist the model in learning the interclass differences of various buildings. SBA-Net [40] integrates two closely related tasks of boundary prediction and SS and designs a plug-and-play attention module to realize the joint learning of the two. Gao et al. [41] explored a joint learning strategy for SS and height estimation and designed novel objective functions based on contrastive learning for optimizing the model. To solve the resolution shift problem of RS images, Cai et al. [42] utilized super-resolution technology to propose a multitask domain adaptation network for SS, improving the model's ability to segment low-resolution small objects.

To sum up, the existing multitask joint learning methods for RS scenes have mainly focused on tasks with strong correlations in the visual field in recent years. There remain possibilities and prospects for cross-modal, heterogeneous data, and potentially related tasks in the field of RS.

III. Methodology

In this section, we propose SIRS, a multitask joint learning framework that achieves CIR and segmentation mask prediction simultaneously. Fig. 3 shows its architecture. This section consists of five parts: problem setup, SSA, AMW, RSITMD-SS annotation, and loss function.

A. Problem Setup

The model input includes images: $\mathcal{I} \in \mathbb{R}^{3\times H\times W}$, text: $\mathcal{T} \in \mathbb{R}^{N}$ and segmentation masks: $\mathcal{M} \in \mathbb{R}^{H\times W}$, which guide the model to learn CIR and SS tasks.

First, we apply the encoders on image and text, respectively,

$$\begin{cases} \mathcal{F}_i = \text{encode}_v(\mathcal{I}) \\ \mathcal{F}_t = \text{encode}_t(\text{preprocess}(\mathcal{T})) \end{cases} \tag{1}$$

where $\mathcal{F}_i = \{f_l\}_{l=1}^{L}$ represents the hierarchical features of the visual encoder output. The encode_v can be the SVP model ResNet series [43] or the CLIP model ViT series [44]. The preprocess implements tokenization of input text, and the encode_t can be selected from BiGRU [45] and transformers.

Then, given $\mathcal{F}_i$, we input it into a segmentation decoder to obtain prediction results $\hat{\mathcal{P}}_{\text{ss}}$

$$\hat{\mathcal{P}}_{\text{ss}} = \text{decode}_{\text{ss}}(\mathcal{F}_i) \tag{2}$$

where the $\text{decode}_{\text{ss}}$ can be instantiated as any general SS decoder.

B. Semantic-Guided Spatial Attention

1) Problem Analysis: The current goal of the CIR task is to match the entities in the textual query with the corresponding foreground objects in the image. Furthermore, it can achieve the degree of modeling the relative positional relationship of entities and objects between the two modalities in the future. Limited by the image–sentence level annotation granularity of the CIR task dataset, existing methods can only learn the above goals in an unsupervised manner. It undoubtedly increases the difficulty of model learning and significantly reduces learning efficiency. Due to the lack of guidance for the corresponding objects in the image, the background noise (not mentioned in the query) may interfere with the model learning efficiency. It further leads to feature misalignment problems in the similarity calculation process with text modal data.

To alleviate the above feature misalignment problem, we propose a SSA module, as shown in Fig. 4. It contains an SS branch and a BR branch. SS introduces an SS task

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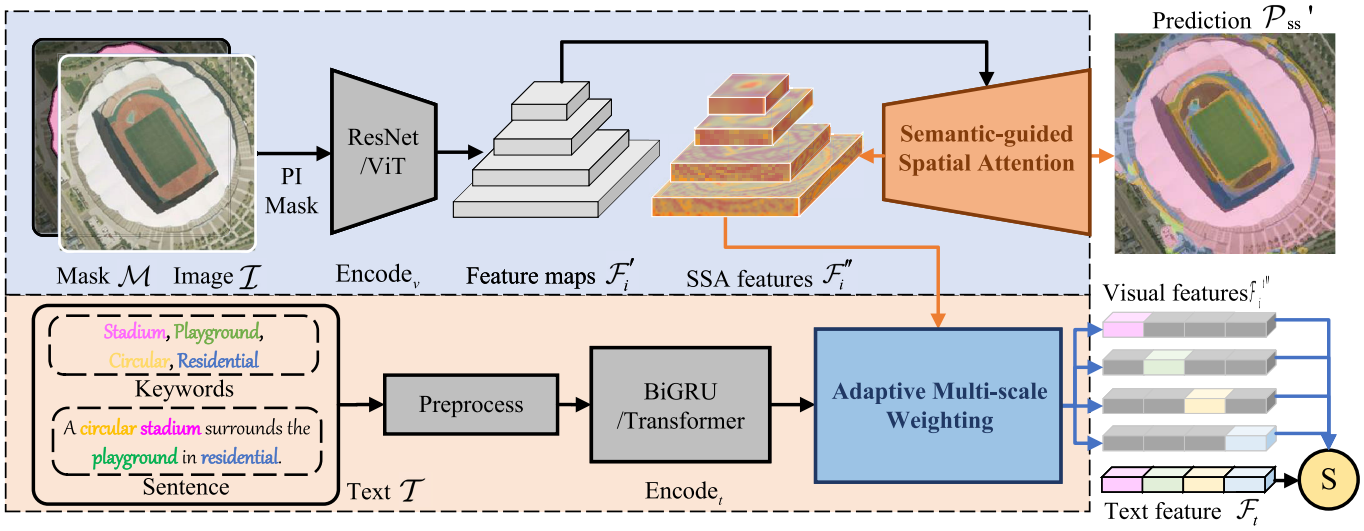


Fig. 3. SIRS architecture takes images, masks, and text as input. Segmentation and similarity results are derived after passing through SSA and AMW, which are both plug-and-play.

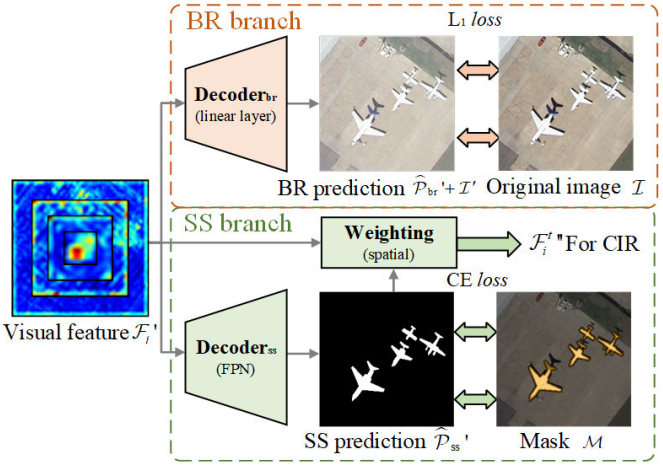


Fig. 4. SSA consists of two parts, namely, the BR branch and the SS branch. The former is to perceive background noise, while the latter learns foreground categories.

learned at the image pixel level to obtain a more fine-grained image feature representation. Furthermore, it utilizes the segmentation results to guide the model to align the image foreground with the text query entities. BR implements the mask reconstruction task based on the background region in the segmentation label. On the one hand, it effectively blocks the noisy interference in the feature alignment process. On the other hand, it guides the model to learn the feature representation of the background region and distinguish it from the foreground features.

2) Mathematical Modeling: In Section III-A, we obtain $\mathcal{F}_i$ output by the visual encoder. From the perspective of the information variety contained, we can further divide it as

$$\mathcal{F}_i = \{\mathcal{F}_{\text{if}}, \mathcal{F}_{\text{ib}}\} \tag{3}$$

where $\mathcal{F}_{\text{if}}$ and $\mathcal{F}_{\text{ib}}$ represent the foreground and background features extracted from the image, respectively. Theoretically, the text feature $\mathcal{F}_t$ is only related to $\mathcal{F}_{\text{if}}$. Therefore, the SS branch aims to guide the model to distinguish the two parts of features in $\mathcal{F}_i$. Furthermore, the BR branch is applied to refine the foreground granularity and weaken the background interference.

BR branch contains a background mask operation and a decoder (linear mapping) for pixel reconstruction, the former as shown below

$$\mathcal{I}' = \text{mask}(\mathcal{I}, \mathcal{M}, \alpha) \tag{4}$$

where α represents the mask ratio, ranging from 0 to 1, and mask represents the PIMask Strategy [1]. The mask operation guides the model to learn the feature representation of the background area and improve the distinction from the foreground. During the whole process, the complete foreground information is retained, reducing the overall input information to increase the attention to the foreground.

The obtained background masked image $\mathcal{I}'$ may replace the original image in (1), and the output is defined as $\mathcal{F}_i'$. Then,

$$\hat{\mathcal{P}}_{\text{br}}' = \text{decode}_{\text{br}}(\mathcal{F}_i') \tag{5}$$

is applied. Since $\mathcal{I} - \mathcal{I}'$ contains the information of the masked area. The goal is to minimize $L_{\text{br}}(\hat{\mathcal{P}}_{\text{br}}', \mathcal{I} - \mathcal{I}') \to 0$, aiming to reconstruct the masked background.

Simultaneously, SS branch inputs $\mathcal{F}_i'$ and performs spatial attention weighting based on $\hat{\mathcal{P}}_{\text{ss}}'$ as

$$\begin{aligned}\mathcal{F}_i'' &= \text{weighting}_s(\mathcal{F}_i', \hat{\mathcal{P}}_{\text{ss}}', \beta) \\ &= n\{m\{\text{ds}[\beta \cdot \hat{\mathcal{P}}_{\text{ss}}' + (1-\beta)\cdot(1-\hat{\mathcal{P}}_{\text{ss}}')]\}\cdot \mathcal{F}_i'\}\end{aligned} \tag{6}$$

where weighting$_s$ contains normalization $n(\cdot)$, mapping $m(\cdot)$, downsampling $\text{ds}(\cdot)$ operators, and β is the weighting factor (0–1), which controls the attention degree to the foreground. See Section IV-D1 for the ablation studies of relevant operators and hyperparameters.

C. Adaptive Multiscale Weighting

1) Problem Analysis: Visual modality possesses a higher dimensional representation space and a more dispersed sample distribution, comparing text data. Currently, the main difficulty in achieving feature alignment between the two modalities lies in the appropriate and efficient visual feature dimensionality

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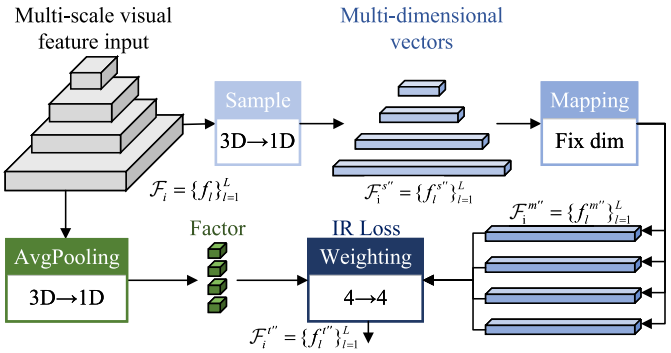


Fig. 5. Module structure of AMW. It is based on the ideas of feature pyramid and intermediate supervision and performs many-to-many similarity calculations with text features.

reduction and extraction. Corresponding to this process is the transformation of visual features in the calculation of feature similarity ($\mathcal{F}_i = \{f_l\}_{l=1}^{L} \rightarrow \mathcal{F}_i^t \in \mathbb{R}^C$). Existing methods usually rely on a single feature map fused from the multiscale outputs ($\mathcal{F}_i^t = f_L$) to calculate similarity with text feature [5], [36], [46], [47], [48]. The one-to-one calculation applied in the above methods is unstable and unbalanced, making it difficult for the model to align the features between vision and text. In addition, there are usually complex backgrounds and various scale objects in RS scenes, which exacerbates the difficulties of the above feature alignment problem.

Inspired by the ideas of intermediate supervision [49] and feature pyramid [50], we expect to fully utilize the feature outputs from different stages of the visual encoder. Information from different layers can be integrated reasonably to adapt to various scale targets in the complex RS scenes. It improves the discriminability and robustness of visual feature learning, especially in the early stages of encoding.

2) Module Design: Fig. 5 shows the module structure of AMW. It takes the multilayer feature maps $\mathcal{F}_i''$ from SSA as input and dynamically integrates them. It mainly includes operations such as sampling, mapping, attention calculation, and weighting. Sampling is applied to compress the feature maps as

$$\begin{aligned}\mathcal{F}_i^{s''} &= \left\{\text{sample}\left(f_l''\right)\right\}_{l=1}^{L} \\ &= \left\{f_l^{s''}\right\}_{l=1}^{L}\end{aligned} \tag{7}$$

where spatial dimensions $H \times W$ are sampled, deriving vector from each layer. The next step is to map the various layer vectors as

$$\begin{aligned}\mathcal{F}_i^{m''} &= \left\{\text{mapping}\left(f_l^{s''}\right)\right\}_{l=1}^{L} \\ &= \left\{f_l^{m''}\right\}_{l=1}^{L}.\end{aligned} \tag{8}$$

The purpose is to map the channels of each layer to the same dimension. Finally, we perform layer-wise weighting on F_i to guide the model to learn the importance of the output features of various layers, i.e.,

$$\begin{aligned}\mathcal{F}_i^{t''} &= \left\{\text{weighting} \cdot f_l^{m''}\right\}_{l=1}^{L} \\ &= \left\{f_l^{t''}\right\}_{l=1}^{L}\end{aligned} \tag{9}$$

where every visual vector $f_l^{t''}$ is with the same dimensions as the text vector $\mathcal{F}_t$. In the next step, they will calculate the similarity with $\mathcal{F}_t$ separately and take the average. For the selection and results of each operator, please refer to Section IV-D2.

D. RSITMD-SS Annotation

1) Motivation: The current RSCIR task mainly faces the following challenges. 1) RS scenes are numerous and contain complex textures, resulting in relatively inconspicuous differences between foreground and background. 2) Foreground objects possess various structural types (rigid, flexible, and combined) and wide-scale ranges. Given the above phenomenon, how to effectively increase intraclass differences, and reduce interclass similarity becomes particularly important. 3) The existing datasets only provide image–sentence granularity annotation, which is not enough.

One of the most intuitive solutions is to guide the model to learn a more fine-grained data distribution. Based on the RSITMD dataset [5], we perform pixel-level mask annotation of some entities involved in the query, named RSITMD-SS. To our best knowledge, it is the first dataset that provides the tasks of both image–text retrieval and SS in RS scenarios. We expect its emergence to have a positive impact on promoting the research of cross-modal, multitasking, and joint learning methods.

2) Characteristics: As the first dataset contains both text captions and segmentation masks, the proposed RSITMD-SS is superior in the following three characteristics.

1) *Complex and Diverse Scenes:* Existing RS SS datasets usually contain few specific scenes. For example, ISPRS Potsdam[1] contains urban street views, RoadNet [51] covers road networks, and GF1_WHU [52] contains various categories of cloud masks. Excessive simplification of the scene will reduce the difficulty of model learning and weaken its generalization ability. Especially under the RSCIR task, introducing more scenes, on the one hand, increases the task complexity. On the other hand, it can help the model learn more diverse RS general feature representations.
   As shown in Fig. 6, the reason we chose the RSITMD dataset for annotation is that it contains a large number of known scenes, far exceeding the current mainstream RS SS datasets.
2) *Categories Based on High-Frequency Text Entities:* Considering the order of frequencies and the recognition of annotations (see Table I), we finally determine the categories applied for SS. They are masks containing 14 categories of foreground objects for SS tasks strictly corresponding to textual captions. The labeling standard depends on whether it coexists in text and image, and is distributed in the dataset widely.
   Fig. 7 indicates that the proportion of each category in the dataset (in image units) is relatively uniform, and the sample quantities of each category are sufficient.

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-sem-label-potsdam/

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Fig. 6. RSITMD-SS covers natural scenes, artificial scenes, regions, transportation-related, and buildings. It can be further refined into 33 scenes, including almost all RS scenes.

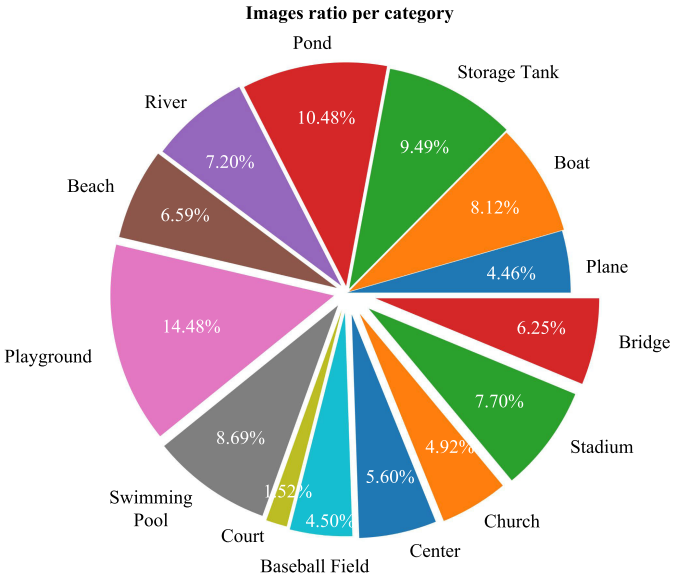


Fig. 7. This pie shows the proportion of images in each category with the SS label.



Fig. 8. (Top) Visualization of partial images and (bottom) corresponding segmentation labels from the IRSS dataset.

The visualization of some images and the corresponding segmentation masks are shown in Fig. 8.

TABLE I

RELATIONSHIP BETWEEN SEMANTIC ANNOTATION CATEGORIES AND TEXT ENTITY FREQUENCIES IN CORRESPONDING SCENARIOS. AND ⋆ INDICATES THE SAME AS THE CATEGORY NAME

Category	Scene	Entity Fequencies		
		No.1	No.2	No.3
Plane	Airport, Plane	⋆	Building	Runway
Bridge	Bridge	⋆	River	Tree
Stadium	Stadium	⋆	Football F.	Building
Church	Church	⋆	Building	Roof
Center	Center	⋆	Building	Tree
Baseball F.	Baseball F.	⋆	Tree	Road
Court	Playground	Playground	Tree	⋆
Swimming P.	Sparse R., Park	House	⋆	Roof
Playground	Playground	⋆	Tree	Court
Beach	Beach	⋆	Sea	Wave
River	River	⋆	Forest	Tree
Pond	Pond	⋆	Tree	House
Storage T.	Storage T.	⋆	Reservoir	Tree
Boat	Port, Boat	Port	⋆	Tree

E. Loss Function

Following existing methods, we apply the classic Pearson correlation coefficient [53] to measure the similarity of visual and text features

$$S\left(\mathcal{F}_i^t,'' \mathcal{F}_t\right) = \sum_{k=1}^{N}\sum_{j=1}^{M} \frac{\mathcal{F}_{i,k}^{t\,\prime\prime} \cdot \mathcal{F}_{t,j}}{\sqrt{\left\|\mathcal{F}_{i,k}^{t\,\prime\prime}\right\|^2} \cdot \sqrt{\left\|\mathcal{F}_{t,j}\right\|^2}} \tag{10}$$

where $N = M$ represents the images and texts number in each batch, respectively. Next, we adopt the classic triplet loss

$$\begin{aligned} L_{\text{cir}}\left(\mathcal{F}_i^t,'' \mathcal{F}_t\right) = &\sum_{\bar{\mathcal{F}}_t}\left[\alpha - S\left(\mathcal{F}_i^t,'' \mathcal{F}_t\right) + S\left(\mathcal{F}_i^t,'' \bar{\mathcal{F}}_t\right)\right]_+ \\ &+ \sum_{\bar{\mathcal{F}}_i^{t\,\prime\prime}}\left[\alpha - S\left(\mathcal{F}_i^t,'' \mathcal{F}_t\right) + S\left(\bar{\mathcal{F}}_i^t,'' \mathcal{F}_t\right)\right]_+ \end{aligned} \tag{11}$$

where α represents the margin, $[\cdot]_+ = \max(\cdot, 0)$, and $\bar{\mathcal{F}}_i^{t\,\prime\prime}$ and $\bar{\mathcal{F}}_t$ stand for the negative samples of image and text modalities, respectively.

For the SS task, the multiclass cross entropy loss is applied

$$L_{\text{ss}}\left(\hat{\mathcal{P}}_{\text{ss}}, \mathcal{M}\right) = \sum_{i=1}^{H\times W} m_i \cdot \log\left(\hat{p}_i\right) \tag{12}$$

where $\hat{p}_i \in \hat{\mathcal{P}}_{\text{ss}}, m_i \in \mathcal{M}$ represent the single pair of prediction pixel and label pixel, respectively.

During training, the overall loss function is

$$L = L_{\text{cir}} + \gamma_1 \cdot L_{\text{ss}} + \gamma_2 \cdot L_{\text{br}} \tag{13}$$

where L_{br} selects the standard L_1 distance, $\gamma_1 = 4$, and $\gamma_2 = 0.5$. Please refer to Section IV-D3 for related results and analysis.

IV. EXPERIMENTS

In this section, we first introduce the dataset applied and the implementation details in Section IV-A. Section IV-B introduces the evaluation metrics. The model comparison and module ablation are described in Sections IV-C and IV-D. We show the visualization results and analysis in the last Section IV-E.

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A. Datasets and Implementation Details

1) Datasets: We test the performance of the proposed SIRS on three datasets. Among them, the first two are open-source mainstream CIR datasets. The last is derived by adding SS masks to the open-source RSITMD dataset, namely, RSITMD-SS.

1) UCM [26] covers urban areas from the United States Geological Survey (USGS) national map. It contains 2100 images of 256 × 256 pixels, covering 21 scenes. Each image corresponds to five captions, with a textual redundancy problem [5]. It may cause high caption similarity between sentences corresponding to different images, increasing the difficulty for the model to learn the samples' distinction. Several classical methods have been tested and compared on it.
2) RSICD [27] is the largest commonly used dataset for the RS CIR task. It contains 10 921 images of 224 × 224 pixels, covering 30 scenes. Each image also corresponds to five captions, which also possesses the problem of redundant text captions. It helps the models verify their generalization and robustness with its relatively large data and scene diversity.
3) RSITMD-SS is derived by adding segmentation masks to the RSITMD [5] dataset. It contains 4743 images of 256 × 256 pixels, covering 33 scenes. We choose RSITMD for its diversities of textual captions and coverage scenarios. Specifically, the five captions corresponding to each image are labeled more granularly, improving the difference among samples. On this basis, we introduce 14 categories of image foregrounds that appear frequently in the captions and label SS masks on them. As the first dataset with foreground-entity level granularity in the RSCIR task, it brings new vitality to researchers in this field.

2) Implementation Details: We use a single NVIDIA GeForce RTX 3090 graphics processing unit (GPU) with 24 GB of memory for model training and testing. The deep learning framework selected is Pytorch 1.7.0. We apply the Adam optimizer with a learning rate of 0.002. The learning rate decay strategy is to change to the original 70% every 20 epochs.

Data augmentation and splitting are consistent with [5], aiming to ensure fairness in comparison. They are 80% for training, 10% for validation, and 10% for testing. Meanwhile, in order to stabilize the model's prediction results, we adopt a ten-fold verification. Among them, the result of each fold takes the average of its last 10 epochs (total epoch 35). It can effectively reduce the problem of variance in model predictions caused by random data splitting (as shown in Fig. 9).

Our SIRS is based on feature encoders of SVP and CLIP methods. Specifically, ResNet-18 [43]/BiGRU [45] for SVP and ViT-B-32 [44]/Transformer for CLIP. During the calculation of feature similarity, except for [30], [31], [54], [55], the other methods all utilize 512 dimensions and achieve better performance. To make a fair comparison with these popular methods, we also choose 512 dimensions. For the SS task, we adopt a simplified Feature Pyramid Network (FPN) [56] as the decoder part.

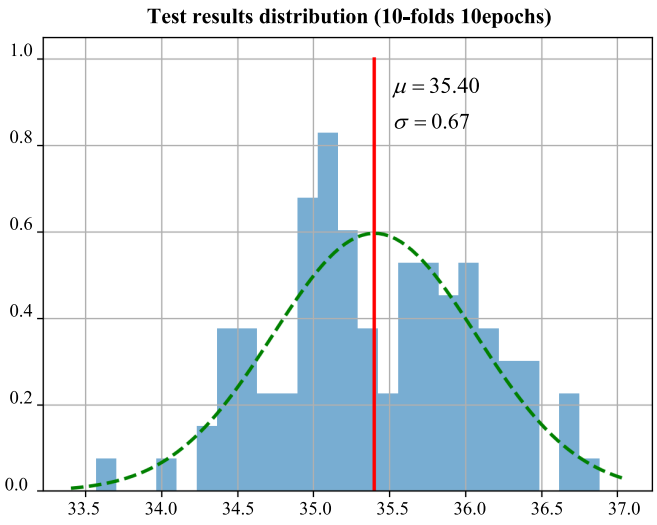


Fig. 9. Distribution of prediction results of the model under the ten-fold 10-epoch average strategy.

B. Evaluation Metrics

This section consists of retrieval and segmentation accuracy metrics, and the model efficiency metrics. They are used for comprehensive model performance evaluation.

1) Retrieval Metrics:

1) Recall at K (**R@K**) is used to evaluate the accuracy of the retrieval model. K represents the number of samples selected for matching, usually $K = 1, 5, 10$. Recall can be calculated as

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

where TP represents the number of true positive samples, and FN represents the number of false negative samples.
2) Mean recall (**mR**) is also introduced to balance the retrieval accuracy. It represents the average of six R@K values (image retrieval text, and reverse).

2) Segmentation Metrics:

1) *Precision* is a basic metric of model performance. The calculation is as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

where FP represents the number of false positive samples.
2) Overall accuracy (**OA**) represents global accuracy. It does not consider categories, only how well all samples are classified

$$\text{OA} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{16}$$

3) Mean intersection over union (**mIoU**) is a standard model accuracy metric in SS tasks. It refers to the intersection over the union of the predictions and the ground truth set

$$\text{mIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}. \tag{17}$$

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TABLE II

COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE TEST SET OF RSITMD-SS. FOR THE SVP METHOD, ALL PARAMETERS ARE LEARNABLE. FOR THE CLIP METHOD, THE PARAMETERS ARE LISTED IN THE FORM OF LEARNABLE/TOTAL

Method	Backbone (image/text)	T2I Result	Sentence Retrieval			Image Retrieval			Evaluation Metrics		
			R@1	R@5	R@10	R@1	R@5	R@10	mR	Params	RT
Supervised vision-pretrained methods											
VSE++ [30]	Resnet-18/GRU	Image	10.38	27.65	39.60	7.79	24.87	38.67	24.83	15.78	18.74
SCAN i2t [31]	Resnet-18/BiGRU	Image	11.06	25.88	39.38	9.82	29.38	42.12	26.28	13.68	49.15
SCAN t2i [31]	Resnet-18/BiGRU	Image	10.18	28.53	38.49	10.10	28.98	43.53	26.64	13.68	49.15
CAMP-bce [54]	Resnet-18/BiGRU	Image	9.07	23.01	33.19	5.22	23.32	38.36	22.03	27.03	35.76
CAMP-triplet [54]	Resnet-18/BiGRU	Image	11.73	26.99	38.05	8.27	27.79	44.34	26.20	27.03	35.76
MTFN [55]	Resnet-18/BiGRU	Image	10.40	27.65	36.28	9.96	31.37	45.84	26.92	77.90	35.97
LW-MCR-b [46]	SqueezeNet/-	Image	9.07	22.79	38.05	6.11	27.74	49.56	25.55	**1.65**	**18.25**
LW-MCR-d [46]	SqueezeNet/-	Image	10.18	28.98	39.82	7.79	30.18	49.78	27.79	**1.65**	**18.25**
LW-MCR-u [46]	SqueezeNet/-	Image	9.73	26.77	37.61	9.25	34.07	54.03	28.58	**1.65**	**18.25**
AMFMN [5]	Resnet-18/BiGRU	Image	11.06	24.78	41.81	11.51	34.69	54.87	29.72	35.94	36.68
MCRN [47]	Resnet-18/BiGRU	Image	13.27	29.42	41.59	9.42	35.53	52.74	30.33	52.35	57.10
GaLR [36]	Resnet-18/BiGRU	Image	13.05	30.09	42.70	10.47	36.34	53.35	31.00	46.89	47.97
GaLR-MR [36]	Resnet-18/BiGRU	Image	14.82	31.64	42.48	11.15	36.68	51.68	31.41	46.89	47.97
baseline [5]	Resnet-18/BiGRU	Image	10.38	27.65	39.60	7.79	24.87	38.67	24.83	33.86	25.81
SIRS w.o. SSA	Resnet-18/BiGRU	Image	14.31	30.39	41.89	10.98	36.53	58.92	32.17	34.11	30.15
SIRS (Ours)	Resnet-18/BiGRU	**Image+Mask**	**15.25**	**35.01**	**48.73**	**11.98**	**41.29**	**60.15**	**35.40**	40.75	40.66
Contrastive vision-language-pretrained methods											
CoOp [57]	ViT-B-32/Transformer	Image	12.19	30.69	42.82	9.16	33.85	54.35	30.51	**0.10/143.85**	-
VPT [58]	ViT-B-32/Transformer	Image	14.98	32.05	40.15	15.97	41.35	60.35	34.14	0.46/144.21	-
VL-Prompt	ViT-B-32/Transformer	Image	12.81	31.28	42.64	12.61	36.20	58.84	32.40	0.47/144.22	-
Adapter [59]	ViT-B-32/Transformer	Image	13.75	27.64	39.96	12.89	40.09	59.91	32.37	0.17/143.92	-
Convpass [60]	ViT-B-32/Transformer	Image	16.03	30.16	40.26	12.05	38.66	58.11	32.55	0.14/143.89	64.74
CLIP-Adapter [48]	ViT-B-32/Transformer	Image	12.83	28.84	39.05	13.30	40.20	**60.06**	32.38	0.52/144.27	27.49
SIRS w.o. SSA	ViT-B-32/Transformer	Image	14.43	43.21	56.22	16.80	38.29	55.96	37.49	1.48/145.23	31.76
SIRS (Ours)	ViT-B-32/Transformer	**Image+Mask**	**18.45**	**44.06**	**57.73**	**19.44**	**43.18**	58.40	**40.21**	8.12/151.87	41.61

Especially, frequency weighted intersection over union (**FWIoU**) weights the proportion of samples based on all categories of IoU.

4) F_1-score represents the harmonic mean of precision and recall. It is calculated as follows:

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \tag{18}$$

3) Efficiency Metrics:

1) Params indicates the number of parameters (unit: MB) applied during model training. It describes the model size and is used to reflect the needs for model storage.
2) Retrieval time (RT) stands for the total retrieved time (unit: second) per epoch. It is usually related to the data size, batch size, and the model inference speed. To be fair, we apply the same batch as 64 for RT testing under the same data.
3) FLOPs indicates the number of floating point operations, which is used to measure algorithm/model complexity.

C. Comparisons With the State-of-the-Art Approaches

In this section, we compare our SIRS with the mainstream methods in the RSCIR task. They are the SVP and CLIP methods. The proposed SIRS is plug-and-play on most of them. As a result, AMFMN and CLIP-Adapter are selected as baseline models, respectively. Tables II and III show the comparison results, followed by analysis and conclusions.

1) Compared Methods:

1) *SVP methods* usually do not share feature encoders. In other words, it utilizes different encoders to extract features of the two modalities. Specifically, the visual feature encoder adopts a pretraining strategy of supervised learning and only learns on itself modality data. Referring to the experimental results in the previous literature, we organize the SVP methods in Tables II and III. They are VSE++ [30], SCAN [31], CAMP [54], MTFN [55], LW-MCR [46], AMFMN [5], MCRN [47], CABIR [61], and GaLR [36]. The experimental results correspond to UCM [26], RSICD [27], and RSITMD [5] datasets, respectively.
2) *CLIP methods* generally migrate knowledge from learning the text and visual general feature representation. Specifically, it adopts a contrastive learning method to pretrain the encoder on image and text data simultaneously. It can effectively align the two modalities and provide richer prior knowledge for the following tasks. While its shared feature encoder parameters are usually large, increasing the computational cost. We mainly compare prompt learning (CoOp [57], VPT [58], and VL-Prompt) and adapter tuning methods (Adapter [59], Convpass [60], and CLIP-Adapter [48]). Specifically, VL-Prompt is applied according to Jiang et al. [62] on both the visual and text tokens. CLIP-Adapter [48] employs an additional bottleneck layer to build residual connections and learn latent features.

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TABLE III
COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE TEST SET OF RSICD AND UCM

Method	RSICD							UCM						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
Supervised vision pretrained methods														
VSE++ [30]	3.38	9.51	17.46	2.82	11.32	18.10	10.43	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN i2t [31]	5.85	12.89	19.84	3.71	16.40	26.73	14.23	12.85	47.14	69.52	12.48	46.86	71.71	43.43
SCAN t2i [31]	4.39	10.90	17.64	3.91	16.20	26.49	13.25	14.29	45.71	67.62	12.76	50.38	77.24	44.67
CAMP-bce [54]	4.20	10.24	15.45	2.72	12.76	22.89	11.38	14.76	46.19	67.62	11.71	47.24	76.00	43.92
CAMP-triplet [54]	5.12	12.89	21.12	4.15	15.23	27.81	14.39	10.95	44.29	65.71	9.90	46.19	76.29	42.22
MTFN [55]	5.02	12.52	19.74	4.90	17.17	29.49	14.81	10.47	47.62	64.29	14.19	52.38	78.95	44.65
LW-MCR-b [46]	4.57	13.71	20.11	4.02	16.47	28.23	14.52	12.38	43.81	59.52	12.00	46.38	72.48	41.10
LW-MCR-d [46]	3.29	12.52	19.93	4.66	17.51	30.02	14.66	15.24	**51.90**	62.86	11.90	50.95	75.24	44.68
LW-MCR-u [46]	4.39	13.35	20.29	4.30	18.85	32.34	15.59	**18.10**	47.14	63.81	13.14	50.38	79.52	45.35
MCRN [47]	6.59	19.40	30.28	5.03	19.38	32.99	18.95	-	-	-	-	-	-	-
CABIR [61]	**8.59**	16.27	24.13	5.42	**20.77**	33.58	18.12	15.17	45.71	**72.85**	12.67	54.19	**89.23**	**48.30**
GaLR [36]	6.50	18.91	29.70	5.11	19.57	31.92	18.62	-	-	-	-	-	-	-
GaLR-MR [36]	6.59	19.85	**31.04**	4.69	19.48	32.13	18.96	-	-	-	-	-	-	-
AMFMN [5]	5.21	14.72	21.57	4.08	17.00	30.60	15.53	14.76	49.52	68.10	13.43	51.81	76.48	45.68
SIRS w.o. SSA (Ours)	7.34	**20.00**	30.78	**5.76**	20.50	**33.78**	**19.69**	14.50	48.40	72.51	**14.28**	**56.05**	82.58	48.05
Contrastive vision-language pretrained methods														
CoOp [57]	6.32	15.89	27.60	4.31	17.89	31.73	17.29	7.19	42.76	74.19	9.78	48.34	87.41	44.95
VPT [58]	7.23	19.40	30.77	6.94	25.26	40.73	21.72	8.86	47.81	78.62	13.64	55.61	94.05	49.77
VL-Prompt	6.47	17.05	28.68	6.69	22.62	36.16	19.61	7.97	45.48	75.96	12.51	51.57	91.52	47.50
Adapter [59]	8.73	24.73	37.81	8.43	26.02	43.33	24.84	10.17	49.73	**80.08**	**18.20**	**62.47**	**95.77**	**52.74**
Convpass [60]	6.54	19.67	32.78	7.03	23.08	39.15	21.38	**16.46**	54.79	78.78	14.24	58.67	94.51	52.91
CLIP-Adapter [48]	7.11	19.48	31.01	7.67	24.87	39.73	21.65	9.83	42.52	66.79	14.61	51.03	84.14	44.82
SIRS w.o. SSA (Ours)	**10.34**	**27.69**	**39.66**	**8.74**	**30.38**	**49.50**	**27.71**	15.21	**50.89**	76.62	14.63	57.31	89.29	50.66

2) Analysis and Conclusions:

1) *Accuracy Comparison:* Tables II and III show the performance comparison between the proposed SIRS and other methods on the RSITMD, RSICD, and UCM datasets. We draw the following conclusions based on these two. 1) On the RSITMD dataset, SIRS exceeded the baseline method by 6.12/7.83 mR, and on RSICD and UCM by 4.16/6.06 and 2.57/5.84, with significant performance improvements. It reveals that the method has achieved advantages in the RSCIR task and owns strong generalization capabilities between SVP/CLIP methods and different datasets. 2) In the absence of SS annotation (w.o. SSA), SIRS still shows sufficient competitiveness (outperforming other methods in RSITMD and RSICD). It is sufficient to illustrate the effectiveness of the proposed AMW in learning complex foreground features of RS scenes. 3) After the introduction of the SS task, the performance of SIRS has been further improved. In addition, it locates the foreground categories associated with the query based on the predicted mask. The image–text granularity in CIR tasks is further refined to the foreground-entity level, which is unavailable for all other methods.
2) *Efficiency Comparison:* We also compare the performance of all methods in retrieval efficiency in Table II. The evaluation metrics are Params and RT. We can see: 1) in learnable Params, CLIP has significant advantages over SVP, mainly due to the efficient parameter transferring. SIRS has an average increase of 6.21M parameters compared to the baseline model, which mainly comes from the SS decoder in the inference stage. It can be reduced by replacing the decoding header with a lighter version. The CLIP-based methods are three to four times larger than the SVP-based methods in terms of total parameters. Due to the heavier encoder, they require more memory and computational costs. 2) We compare the RT of SIRS with other methods on the RSITMD test set. Without SSA, RT is improved on average by 4.31 s compared to the baseline, which is only slower than VSE++ and the lightweight architecture LW-MCR. The introduction of the SSA module results in an average increase in RT of 10.18 s, which is still faster than MCRN, GaLR, Convpass, and other methods. It is worth mentioning that image segmentation predictions are also output to provide finer retrieval results for queries.

D. Ablation Studies

We conduct ablation experiments on the proposed SSA, AMW, complete loss optimization, and the effect of feature alignment.

1) Semantic-Guided Spatial Attention: Since SSA includes the BR and SS branches, ablation experiments are also conducted on the above two parts, respectively.

For the BR branch, we conduct three experiments, namely, the adjustment of mask ratio α, the influence of L_{br} loss weighted by γ_2, and the effectiveness of the background area masking.

1) α adjusts the proportion of the image background masking area. On the one hand, it reduces background information interference and improves the ability of the

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TABLE IV

ABLATIONS OF SPATIAL ATTENTION WEIGHTING OF SSA ON THE RSITMD-SS DATASET

SS dec.	SSA Map.	SSA DS	SSA Norm.	Metrics mR	Metrics Par.	Metrics FLOPs
with	-	-	-	32.17	1.08	1.28G
with	step	interpolate	-	-4.00	0	0
with	step	avgpool	-	-2.43	0	+0.27M
with	step	maxpool	-	-1.48	0	0
with	identity	interpolate	-	-0.37	0	0
with	identity	avgpool	-	+1.50	0	+0.27M
with	identity	maxpool	-	+1.81	0	0
with	identity	avgpool	BN	+2.10	≈0	+2.23M
with	identity	avgpool	GN	+2.62	0	+0.27M
with	identity	maxpool	BN	+2.24	≈0	+1.97M
with	**identity**	**maxpool**	**GN**	**+3.23**	**0**	**0**

TABLE V

TUNING OF α AND β ON THE RSITMD-SS DATASET TO VERIFY THE MR EVALUATION METRIC

α	**0.05**	**0.10**	**0.15**	**0.20**	**0.30**	**0.40**	**0.50**
mR	35.79	36.97	**37.16**	35.87	34.80	29.78	18.34
β	**0.30**	**0.40**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**
mR	15.67	24.21	33.97	35.79	**37.16**	37.03	36.70

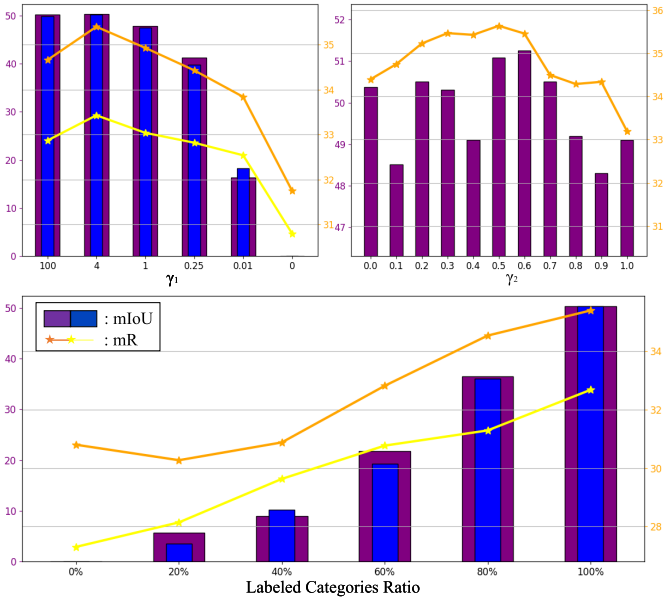


Fig. 10. (Top left) Ablation of the γ_1 (ratio of L_{ss} loss to the total loss). (Top right) Ablation of the γ_2 (ratio of L_{br} loss to the total loss). (Bottom) Ablation of the segmentation labels proportion in the total data. Purple/orange colors represent that spatial attention is applied, and blue/yellow represents the opposite.

model to perceive it. On the other hand, it expands the variance between training and test data. We set $\beta = 0.6$ before adjusting α. As Table V shows, the larger the α value, the lower the background feature interference, but the greater the difference between training and test data. The experimental results verify our hypothesis. α improves performance within a small value range (0–0.2), and as the value increases, performance drops sharply.

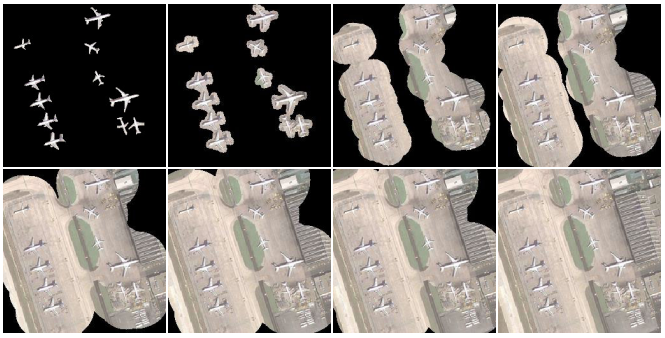


Fig. 11. (From left to right, top to bottom) Sample of the unlabeled area coverage gradually expanded.

TABLE VI

RELATIONSHIP BETWEEN THE COVERAGE RATIO OF UNLABELED AREAS AND RETRIEVAL ACCURACY IN THE RSITMD-SS

Ratio(%)	**0**	**15**	**30**	**45**	**60**	**75**	**90**	**100**
mR	33.1	35.9	37.3	39.8	**41.6**	40.7	40.3	40.2

2) γ_2 controls the proportion of the BR branch in the total loss. It is shown in the top left of Fig. 10, as γ_2 increases, the model performance shows a trend of increasing and then decreasing. It demonstrates that as the model continues to learn the feature representation of the background information, it helps to distinguish the features between the foreground and background to a certain extent, illustrating the effectiveness of the BR branch.
3) To further verify the positive impact of the BR branch, we conduct another experiment. As shown in Fig. 11, based on SS annotation, we cover unlabeled areas on the complete dataset (including the training and test set). Table VI shows that as the recovered ratio of the background gradually increases, the model performance shows a trend of sharp increase and then slowly decrease. It reveals that appropriate background area masking can effectively improve the model performance. It is worth noting that complete coverage has a significant impact on model performance, which could be due to the misalignment between the labeled categories and the corresponding text. For example, the lawn mentioned in the airport is not annotated in the SS label.

For the SS branch, we mainly verify the selection of hyperparameter β and several operators.

1) β sets different levels of attention to foreground and background features in the spatial attention weighting process. Theoretically, when $\beta > 0.5$, it suppresses the background and enhances the foreground information; when $\beta \leq 0.5$, the opposite is true. In Table V, we set $\alpha = 0.15$ before adjusting β. Experimental results show that when $\beta \leq 0.5$, it has a greater impact on model performance. As β continues to increase, mR displays a trend of first increasing and then decreasing. We deduce that it may be due to the difference between the labeled segmentation foreground and corresponding text entities. For example, entities such as runways, lawns, and ter-

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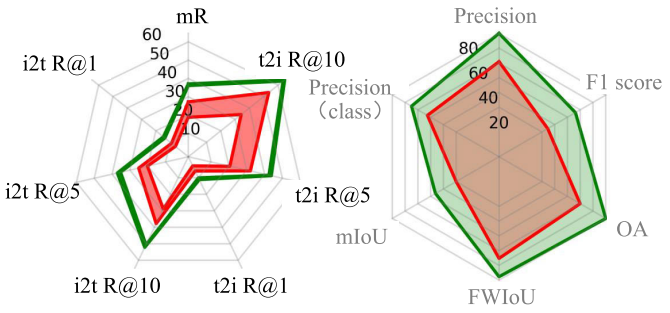


Fig. 12. Comparison of model performance on CIR (left) and SS (right) tasks before (red) and after (green) applying AMW. Precision (class) represents the average accuracy of various categories.

minals mentioned in the text have not yet been labeled in the segmentation categories.

2) Regarding the selection of normalization, downsampling, and mapping operators in (6), we provide various solutions, as shown in Table IV. Normalization includes batch and group normalization, downsampling includes linear interpolation, average pooling, and max pooling, and mapping includes step and identity mapping. As seen from Table IV, the overall performance ranges between −4.00 and +3.23 mR and the best balance of accuracy and efficiency is under identity mapping, max pooling, and group normalization.

2) Adaptive Multiscale Weighting: The ablations mainly include hyperparameter tuning and overall comparison.

Regarding the hyperparameter tuning, we mainly select operators for the three parts of the module composition: sampling, mapping, and weighting. Sampling includes median, maximum, and mean. Mapping includes interpolation, convolution, and fully connected (FC). Average and adaptive average pooling (AAP) operations are optional. The baseline is a model without an SS branch and AMW module.

As shown in Table VII, the model accuracy range fluctuates between 10 mR, and the parameters and FPS fluctuate slightly. We choose the last line to configure AMW, under acceptable computing resource consumption and ideal retrieval accuracy.

After completing the hyperparameter selection of AMW, we ablate the overall module. As shown in Fig. 12, the left corresponds to the performance improvement of AMW on the CIR task. It is worth mentioning that the colored range between the lines in the graph represents the fluctuation of the prediction results. Based on the model training strategy in Section IV-A2, we count the ten-fold results and calculate the mean and standard deviation, thereby obtaining the fluctuation range. It illustrates the stability of the model on the corresponding dataset. The right corresponds to the performance comparison of the SS task. AMW has achieved steady improvement in all six evaluation metrics. In summary, AMW has improved the model performance in terms of accuracy and robustness on two tasks, due to its effective learning of multiscale feature map information.

3) Complete Loss Optimization: For the complete loss function, we have designed two sets of experiments focusing on the γ_1 and γ_2 tuning, respectively. They explore the optimal convergence strategy of the multitask model (see Fig. 10 top).

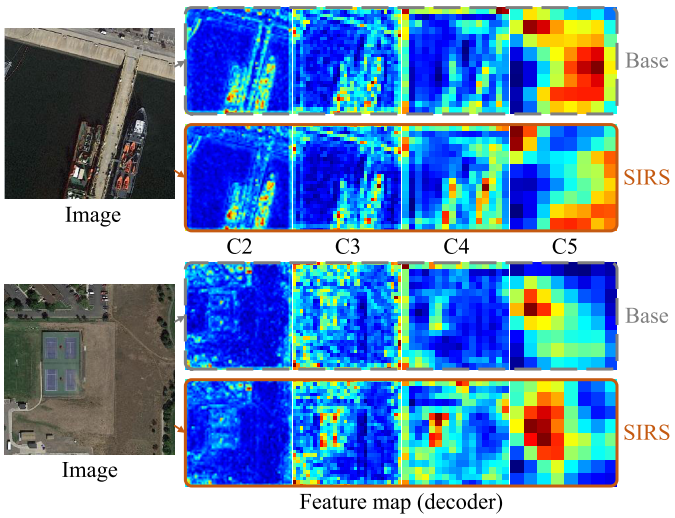


Fig. 13. Feature map visualization of baseline and the proposed SIRS.

The experimental conclusions are summarized as follows.

1) In a wide range greater than 1, γ_1 has a tiny impact on mIoU and exhibits convex changes on mR (optimal when $\gamma_1 = 4$). As γ_1 continues to decrease below 1, both mR and mIoU reduce rapidly. It shows that applying more weight to the SS branch may further improve the upper limit of retrieval performance. 2) As γ_2 goes from 0 to 1, mR shows convex changes (optimal when $\gamma_2 = 0.5$), while the mIoU changes irregularly. We analyze that if the noise perception branch gains more attention, the background distribution learning will improve. It enlarges the difference with the foreground features, improving the retrieval accuracy. However, if it further increases, the learning of other branches may be affected inevitably, leading to overestimating focus.

4) Analysis of Features Alignment: In this section, we conduct two sets of quantitative experiments on the impact of RSITMD-SS labels on retrieval performance. It mainly includes the impact of the labeled data amount and the coverage scope of unlabeled areas on retrieval accuracy from the SS labels.

As Fig. 10 bottom shows, with the proportion of labeled data increases, mR and mIoU show a steady upward trend. It indicates that when the SS label categories and text entities are relatively correlated, fine-grained learning of visual features can effectively improve the performance upper limit of retrieval tasks.

Fig. 11 shows an example of the results after covering a certain proportion of the unlabeled area in the RSITMD-SS dataset. According to different coverage ratios, eight groups are listed. In each group, the model is trained and validated on both the covered datasets (see Table VI). When the coverage ratio is 60%, the mR is optimal and decreases toward both sides. It reveals that covering unlabeled areas to a certain extent can effectively reduce the noise information in visual features, thereby improving the upper limit of retrieval performance. The coverage level usually depends on the correlation between the SS label categories and the text entities. For example, in Fig. 11, the lawns and terminals contained in the text entities are not labeled. Therefore, if the coverage area is large, the above information may be missing, reducing the retrieval similarity.

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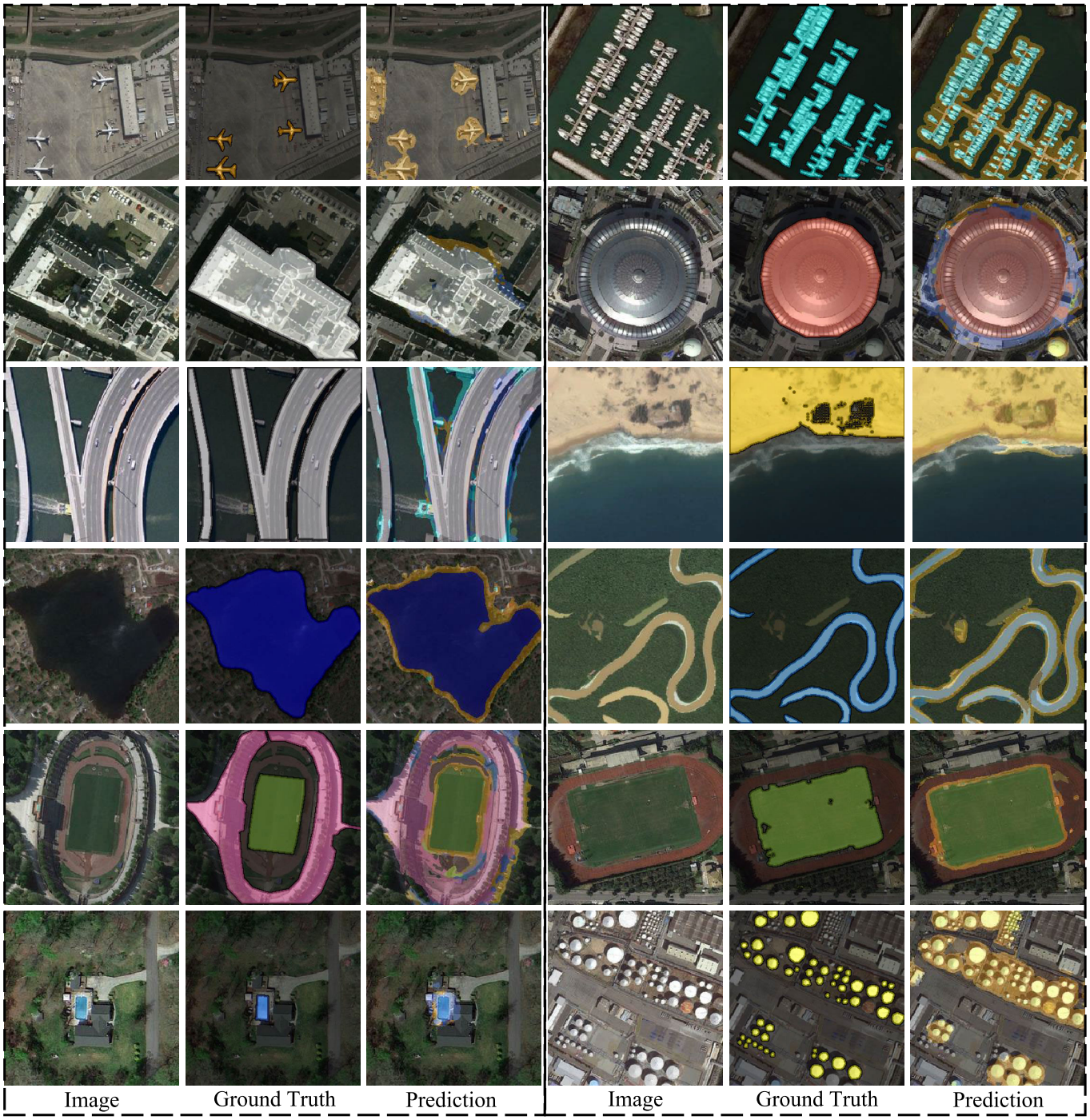


Fig. 14. SIRS segmentation prediction results on the RSITMD-SS test set. Each color represents the same category.

E. Visualization Analysis

In this section, we visualize relevant content in the model inference process, including hierarchical feature maps, segmentation results, and retrieval results.

1) Hierarchical Feature Maps: To qualitatively analyze the feature representations learned by the proposed SIRS at each layer, we visualize the hierarchical feature maps (layer $c_2 - c_5$) output from the encoder. Specifically, we calculate the average of the channel dimension from each feature map level and present them as heat maps. The two models participating in the comparison are SIRS and the baseline model without AMW and SSA modules. Except for model differences, the other variables controlled are the same, including data preprocessing and implementation details.

As shown in Fig. 13, the input images are from the RSITMD test set, including ships and tennis courts, respectively. It reveals, 1) overall, the deeper the feature map layer goes, the lower the resolution and higher the variance can be; 2) between the two models, SIRS distinguishes foreground targets better than the baseline. Specifically, the numerical difference in the foreground position becomes larger and the relative position is getting more accurate; Based on the above phenomenon, we analyze that, 1) as the model layer goes

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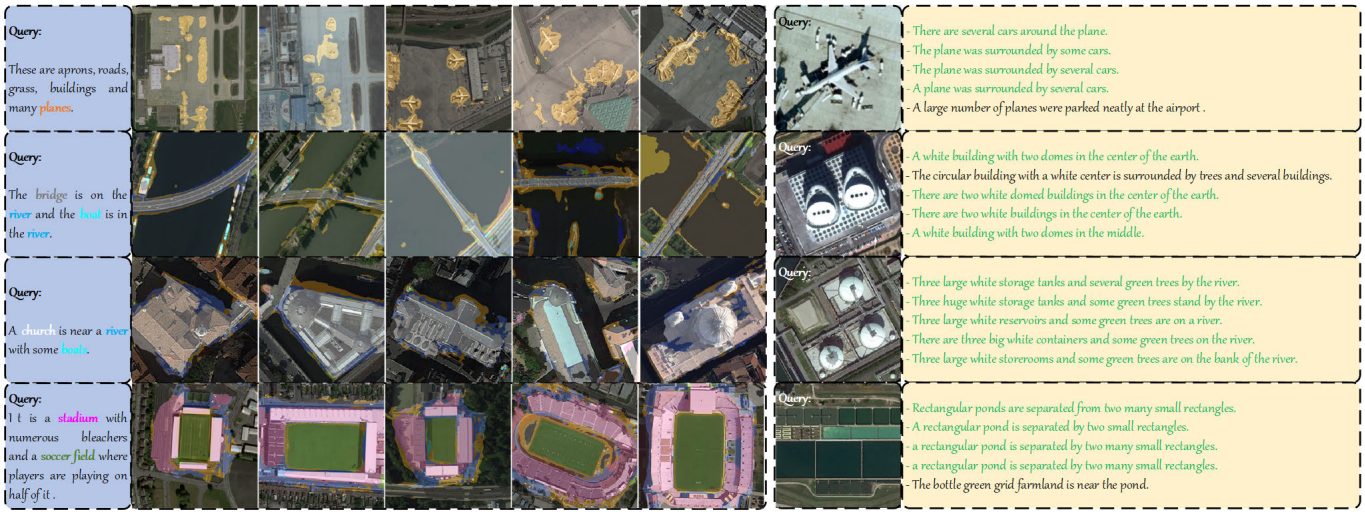


Fig. 15. Image–text and text–image retrieval results of SIRS on the RSITMD-SS test set. In text–image retrieval, each color represents a category. In image–text retrieval, the green shows correctly retrieved sentences.

TABLE VII
ABLATIONS OF THE AMW MODULE ON THE RSITMD-SS DATASET

SSA	AMW			Metrics		
	Sam.	Map.	Wei.	mR	Params	FLOPs
w.o.	-	-	-	28.45	0	2.12G
w.o.	med.	int.	ave.	-4.44	0	+2.79M
w.o.	med.	c1d	ave.	+0.20	+1.41M	+251.66M
w.o.	med.	FC	ave.	+1.19	+0.47M	+0.49M
w.o.	max	int.	ave.	-4.49	0	+2.79M
w.o.	max	c1d	ave.	-2.25	+1.41M	+251.66M
w.o.	max	FC	ave.	+2.33	+0.47M	+0.49M
w.o.	mean	int.	ave.	-2.25	0	+2.79M
w.o.	mean	c1d	ave.	+2.32	+1.41M	+251.66M
w.o.	mean	FC	ave.	+2.35	+0.47M	+0.49M
w.o.	mean	FC	AAP	+3.72	+0.49M	+1.47M
with	**mean**	**FC**	**AAP**	**+6.95**	**+1.57M**	**+1.28G**

deeper, the feature map is downsampled and derives more semantic information. 2) With the help of the SS branch, the encoder learns more accurate positions and feature differences between foreground and background.

2) Segmentation Results: We visualize the prediction results of SIRS on the SS task (see Fig. 14). The categories include small-scale objects, such as planes, ships, storage tanks, and swimming pools. Large-scale objects include churches, centers, ponds, stadiums, and playgrounds. Also included is a class of extreme aspect ratios, like bridges and rivers. Each color represents the same category.

It reveals that SIRS locates small-scale objects accurately, but has poor behavior on edge positions. This is mainly due to the category imbalance caused by the small size and quantity, which influence the performance of these categories. The segmentation results of SIRS at large scales and extreme aspect ratios are stable and excellent. On the one hand, it reflects the completeness of the dataset segmentation mask annotation. On the other hand, it also demonstrates the effectiveness of the model in learning various categories.

3) Retrieval Results: Finally, we show examples of SIRS results in cross-modal image–text and text–image retrieval tasks. Each query corresponds to the top five most confident results. Thanks to the introduction of the SS branch, when performing text–image queries, in addition to images, SIRS can also output prediction masks for various foreground categories. All examples are from the RSITMD dataset test set.

As shown in Fig. 15, in text–image retrieval, the entities corresponding to the text query are basically consistent with the foreground from the predicted mask. It strongly proves that the proposed SIRS establishes entity-foreground level associations in cross-modal retrieval tasks. Mask representation obtains more fine-grained and diverse results, providing a strong basis for retrieval results.

V. CONCLUSION

This article explored a novel plug-and-play cross-modal framework based on multitask joint learning, aiming to refine the RS scene CIR task from the image–sentence level to the foreground-entity level to provide accurate and granular representations for retrieval results. Specifically, by proposing RSITMD-SS and SSA, we introduce the SS task into the existing CIR task and implement a multitask joint learning framework. SSA applies noise perception and SS branches to filter background features and refine foreground features, respectively. The proposed AMW effectively improves the learning effect on foreground targets of large-scale ranges in RS scenes. It alleviates the feature mottle problem caused by a large amount of noise in RS scenarios and further improves model stability. The two modules achieve considerable improvement with lower computing resource consumption without modifying the inherent structure. Sufficient and complete experiments verified the joint learning effect of the proposed SIRS in the two tasks. Applying SIRS, both SVP and CLIP-based methods can steadily improve the average 7 mR. Besides, it additionally outputs mask predictions strongly related to the query entities, achieving foreground prediction, which is not available among other methods. We hope that the proposed RSITMD-SS and SIRS can inspire more research to explore the relationship between various tasks and RSCIR and achieve more fine-grained RS visual language representation.

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